

CMT1306M05 High Reliability Reinforced Isolated Sigma-Delta Modulator

Features

- Up to 5000Vrms Insulation voltage
- Clock frequency: 5MHz to 21MHz
- $\pm 50\text{mV}$ or $\pm 250\text{mV}$ linear Input Voltage Range
- Excellent DC Performance:
 - Offset Error: $\pm 50\mu\text{V}$ or $\pm 100\mu\text{V}$ (Max)
 - Offset Drift: $\pm 1 \mu\text{V}/^\circ\text{C}$ (Max)
 - Gain Error: $\pm 0.2\%$ (Max)
 - Gain Drift: $\pm 40 \text{ ppm}/^\circ\text{C}$ (Max)
- SNR: 82.5dB or 86dB (Typ)
- High CMTI: $150\text{kV}/\mu\text{s}$ (Typ)
- System-Level Diagnostic Features:
 - AVDD monitoring
 - Input common-mode overvoltage detection
- Operation Temperature: $-40^\circ\text{C} \sim 125^\circ\text{C}$
- AEC-Q100 qualified for automotive applications
- UL recognition: up to 5000Vrms for 1 minute per UL1577
- CQC certification per GB4943.1-2011
- CSA component notice 5A approval IEC60950-1 standard
- DIN VDE V 0884-11:2017-01
- RoHS-Compliant Package: SOW8(300mil)

Application

- Shunt current monitoring
- AC motor controls

- Uninterruptible Power Supply
- Automotive onboard charger
- Power and solar inverters

Description

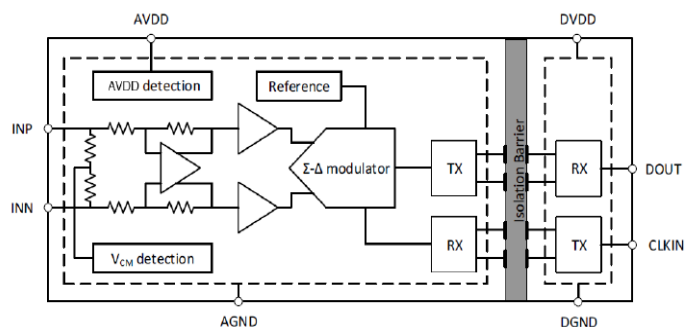
The CMT1306M05 is a high performance Σ - Δ modulator with output separated from input based on the HOPERF capacitive isolation technology. The device has a linear differential input signal range of $\pm 50\text{mV}$ ($\pm 64\text{mV}$ full-scale). The differential input is ideally suited to shunt resistor-based current sensing in high voltage applications where isolation is required.

The analog input is amplified and continuously sampled by a second-order Σ - Δ modulator and converted to a high speed, single bit data stream. The output data is synchronous to the external clock with a frequency range from 5MHz to 21MHz. By using an appropriate digital filter (such as sinc3 filter) to decimate the bitstream, the device can achieve 16 bits resolution and an 86dB/82.5dB signal to noise ratio (SNR) at 78.125kSPS with a 20MHz master clock.

The fail-safe functions including input common-mode overvoltage detection and missing AVDD detection simplify system-level design and diagnostics.

Ordering Information

Part No.	Package	Minimum Order Quantity
CMT1306M05	SOW8 (300mil)	1000



Functional Block Diagram

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1 Specifications

1.1 Recommended Operating Ratings

Over operating free-air temperature range (unless otherwise noted).

Table 1-1. Recommended Operating Ratings

Parameter		Symbol	Min.	Typ.	Max.	Unit
Analog Side Power Supply		AVDD	3.0	5.0	5.5	V
Digital Side Power Supply		DVDD	3.0	3.3	5.5	V
CMT1306M05	Differential input voltage before clipping output	VClipping		±64		mV
	Linear differential input full scale voltage	V _{FSR}	-50		50	mV
	Operating common-mode input voltage	V _{CM}	-0.032		0.8	V
Ambient Temperature		T _A	-40		125	°C

1.2 Absolute Maximum Ratings

Over operating free-air temperature range (unless otherwise noted).

Table 1-2. Absolute Maximum Ratings^[1]

Parameter	Symbol	Min.	Typ.	Max.	Unit
Power Supply Voltage	AVDD, DVDD	-0.3		6.5	V
Analog Input Voltage	INP, INN	AGND-6		AVDD+0.5	V
Digital Input Voltage	CLKIN	DGND-0.5		DVDD+0.5	V
Digital Output Voltage	DOUT	DGND-0.5		DVDD+0.5	V
Output Current per Output Pin	I _O	-10		10	mA
Operating Temperature	T _{OPR}	-40		125	°C
Junction Temperature	T _J	-40		150	°C
Storage Temperature	T _{STG}	-55		150	°C
Electrostatic discharge	HBM ⁽¹⁾	±2000			V
	CDM ⁽²⁾	±1000			V

(1) Human body model (HBM), per AEC-Q100-002-RevD
 (2) Charged device model (CDM), per AEC-Q100-011-RevB

1.3 Thermal Information

Table 1-3. Thermal Information

Parameter	Symbol	CMT1306M05	Unit
		SOW8	
Junction-to-ambient thermal resistance	$R_{\theta JA}$	86	°C/W
Junction-to-case (top) thermal resistance	$R_{\theta JC(top)}$	28	°C/W
Junction-to-board thermal resistance	$R_{\theta JB}$	42	°C/W
Junction-to-top characterization parameter	ψ_{JT}	4	°C/W
Junction-to-board characterization parameter	ψ_{JB}	42	°C/W

1.4 Insulation Specifications

1.4.1 Insulation and Safety Related Specifications

Parameters	Symbol	Value	Unit	Comments
Minimum External Air Gap (Clearance)	CLR	8	mm	Shortest terminal-to-terminal distance through air
Minimum External Tracking (Creepage)	CPG	8	mm	Shortest terminal-to-terminal distance across the package surface
Minimum internal gap	DTI	28	μm	Distance through insulation
Tracking Resistance (Comparative Tracking Index)	CTI	>600	V	DIN EN 60112 (VDE 0303-11); IEC 60112
Material Group		I		IEC 60664-1

1.4.2 Insulation Characteristics

Description	Test Condition	Symbol	Value	Unit
DIN VDE 0110				
Installation Classification per DIN VDE 0110	For Rated Mains Voltage ≤ 150Vrms		I to IV	
	For Rated Mains Voltage ≤ 300Vrms		I to IV	
	For Rated Mains Voltage ≤ 400Vrms		I to IV	
Climatic Classification			40/125/21	
Pollution Degree per DIN VDE 0110, Table 1			2	
Maximum repetitive isolation voltage		V_{IORM}	2121	V_{PEAK}
Maximum working isolation voltage	AC Voltage	V_{IOWM}	1500	V_{RMS}
	DC Voltage		2121	V_{DC}
Input to Output Test Voltage, Method B1	$V_{IORM} \times 1.875 = V_{pd(m)}$, 100% production test, $t_{ini} = t_m = 1$ sec, partial discharge < 5 pC	$V_{pd(m)}$	3977	V_{PEAK}
Input to Output Test Voltage, Method A				
After Environmental Tests Subgroup 1	$V_{IORM} \times 1.6 = V_{pd(m)}$, $t_{ini} = 60$ sec, $t_m = 10$ sec, partial discharge < 5 pC	$V_{pd(m)}$	3394	V_{PEAK}
After Input and /or Safety Test Subgroup 2 and Subgroup 3	$V_{IORM} \times 1.2 = V_{pd(m)}$, $t_{ini} = 60$ sec, $t_m = 10$ sec, partial discharge < 5 pC	$V_{pd(m)}$	2545	V_{PEAK}
Maximum transient isolation voltage	$t = 60$ sec	V_{IOTM}	8000	V_{PEAK}
Maximum Surge Isolation Voltage	Test method per IEC60065,1.2/50us waveform, $V_{TEST} = V_{IOSM} \times 1.6$	V_{IOSM}	6250	V_{PEAK}
Isolation resistance	$V_{IO} = 500V$, $T_{amb} = T_s$	R_{IO}	$>10^9$	Ω
		R_{IO}	$>10^{11}$	Ω

	$V_{IO} = 500V, 100^{\circ}C \leq T_{amb} \leq 125^{\circ}C$			
Isolation capacitance	$f = 1MHz$	C_{IO}	0.8	pF
Safety input, output, or total power	$\theta_{JA} = 86^{\circ}C/W, V_I = 5.5V, T_J = 150^{\circ}C, T_A = 25^{\circ}C$	P_S	1430	mW
Safety input, output, or supply current	$\theta_{JA} = 86^{\circ}C/W, V_I = 5.5V, T_J = 150^{\circ}C, T_A = 25^{\circ}C$	I_S	260	mA
Maximum safety temperature		T_S	150	$^{\circ}C$
UL1577				
Insulation voltage per UL	$V_{TEST} = V_{ISO}, t = 60\text{ s (qualification)}, V_{TEST} = 1.2 \times V_{ISO}, t = 1\text{ s (100% production test)}$	V_{ISO}	5000	V_{RMS}

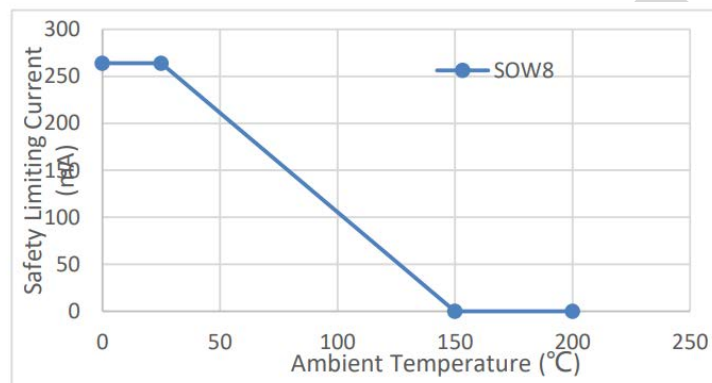


Figure 1-4. CMT1306M05 Thermal Derating Curve, Dependence of Safety Limiting Values
with Case Temperature per DIN VDE V 0884-11

1.5 Safety-Limiting Values

Safety limiting intends to minimize potential damage to the isolation barrier upon failure of input or output circuitry.

Table 1-5. Safety-Limiting Values

Parameter	Symbol	Test conditions	Side	MIN	TYP	MAX	Unit
Safety output supply current	I_S	$\theta_{JA} = 97.3^{\circ}C/W, V_{VDDA/B} = 15\text{ V}, T_J = 150^{\circ}C, T_A = 25^{\circ}C$	DRIVER A, DRIVER B			TBD	mA
		$\theta_{JA} = 97.3^{\circ}C/W, V_{VDDA/B} = 30\text{ V}, T_J = 150^{\circ}C, T_A = 25^{\circ}C$	DRIVER A, DRIVER B			TBD	mA
Safety supply power	P_S	$R\theta_{JA} = 97.3^{\circ}C/W, T_A = 25^{\circ}C, T_J = 150^{\circ}C$	INPUT			TBD	mW
			DRIVER A			TBD	
			DRIVER B			TBD	
			TOTAL			TBD	
Safety temperature ^[1]	T_S					TBD	$^{\circ}C$

Notes:

- [1]. The maximum safety temperature, T_S , has the same value as the maximum junction temperature, T_J , specified for the device. The IS and PS parameters represent the safety current and safety power respectively. The maximum limits of IS and PS should not be exceeded. These limits vary with the ambient temperature, T_A .

The junction-to-air thermal resistance, $R_{\theta JA}$, in the Section 1.4 table is that of a device installed on a high-K test board for leaded surface-mount packages. Use these equations to calculate the value for each parameter:

$T_J = T_A + R_{\theta JA} \times P$, where P is the power dissipated in the device.

$T_{J(max)} = T_S = T_A + R_{\theta JA} \times P_S$, where $T_{J(max)}$ is the maximum allowed junction temperature. $P_S = I_S \times V_I$, where V_I is the maximum input voltage.

1.6 Electrical Characteristics

1.6.1 Electrical Characteristics: CMT1306M05

(AVDD = 3.0V ~ 5.5V, DVDD = 3.0V ~ 5.5V, INP = -50mV to +50mV, and INN = AGND = 0V, $T_A = -40^\circ\text{C}$ to 125°C and sinc3 filter with OSR=256. Unless otherwise noted, Typical values are at CLKIN=20MHz, AVDD = 5V, DVDD = 3.3V, $T_A = 25^\circ\text{C}$)

Table 1-6. Recommended Operating Ratings^[1]

Parameters	Symbol	Min	Typ	Max	Unit	Comments
Power Supply						
Analog Side Supply Voltage	AVDD	3.0	5.0	5.5	V	
Digital Side Supply Voltage	DVDD	3.0	3.3	5.5	V	
Analog Side Supply Current	IAVDD		11.5	15	mA	
Digital Side Supply Current	IDVDD		2	2.5	mA	
AVDD undervoltage detection threshold voltage	AVDD _{UV}	2	2.5	3	V	AVDD falling
Analog Input						
Common-mode overvoltage detection level	VCMov	0.9			V	Detection level has a typical hysteresis of 96 mV
Common-mode rejection ratio	CMRR _{dc}		-95		dB	INP = INN, $f_{IN} = 0\text{ Hz}$, $V_{CM\ min} \leq V_{IN} \leq V_{CM\ max}$
	CMRR _{ac}		-99		dB	INP = INN, $f_{IN} = 10\text{ kHz}$, $V_{CM\ min} \leq V_{IN} \leq V_{CM\ max}$
Single-ended input resistance	R _{IN}		4.75		kΩ	INN = AGND
Differential input resistance	R _{IND}		4.9		kΩ	
Input capacitance	C _I		2	4	pF	
Input bias current	I _{IB}	-24	-23	-20	μA	INP = INN = AGND, $I_{IB} = (I_{IBP} + I_{IBN}) / 2$
Input bias current drift	TCI _{IB}		±2		nA/°C	
Common-mode transient immunity	CMTI	100	150		kV/μs	Common-mode transient immunity

DC Accuracy						
Differential nonlinearity	DNL	-0.99		0.99	LSB	
Integral nonlinearity	INL	-4	±1	4	LSB	
Offset error	E _O	-50	±2.5	50	μV	INP = INN = AGND
Offset error thermal drift	TCE _O	-0.5	±0.15	0.5	μV/°C	
Gain error	E _G	-0.2%	±0.005%	0.2%		
Gain error thermal drift	TCE _G	-30	±10	30	ppm/°C	
Power supply rejection ratio	PSRR		-106		dB	PSRR vs AVDD, at DC
			-104		dB	PSRR vs AVDD, 100mV and 10kHz ripple
AC Accuracy						
Signal to noise ratio	SNR	78	82.5		dB	f _{IN} = 1kHz
Signal to noise and distortion	SINAD	77	82.5		dB	f _{IN} = 1kHz
Total harmonic distortion	THD		-96	-84	dB	f _{IN} = 1kHz
Spurious-free dynamic range	SFDR	86	100		dB	f _{IN} = 1kHz
Digital Input / Output						
Input current	I _{IN}	0		7	μA	DGND ≤ V _{IN} ≤ DVDD
Input capacitance	C _{IN}		5		pF	
Output load capacitance	C _{LOAD}		30		pF	
High-level input voltage	V _{IH}	0.7×DVDD		DVDD+0.3	V	
Low-level input voltage	V _{IL}	-0.3		0.3×DVDD	V	
High-level output voltage	V _{OH}	DVDD-0.1			V	I _{OH} = -20μA
		DVDD-0.4			V	I _{OH} = -4mA
Low-level output voltage	V _{OL}			0.1	V	I _{OL} = 20μA
				0.4	V	I _{OL} = 4mA
Timing						
CLKIN clock frequency	f _{CLKIN}	5		21	MHz	
CLKIN clock period	t _{CLKIN}	47.6		200	ns	
CLKIN clock high time	t _{HIGH}	20	25	120	ns	
CLKIN clock low time	t _{LOW}	20	25	120	ns	
DOUT rising time	t _r			5	ns	C _{LOAD} = 15pF
DOUT falling time	t _f			5	ns	C _{LOAD} = 15pF
DOUT hold time after rising edge of CLKIN	t _H	3.5			ns	C _{LOAD} = 15pF

Rising edge of CLKIN to DOUT valid delay	t_D			15	ns	$C_{LOAD} = 15pF$
Analog setting time	t_{AS}		0.5		ms	AVDD step to 3.0 V with DVDD ≥ 3.0 V, to DOUT valid, 0.1% settling

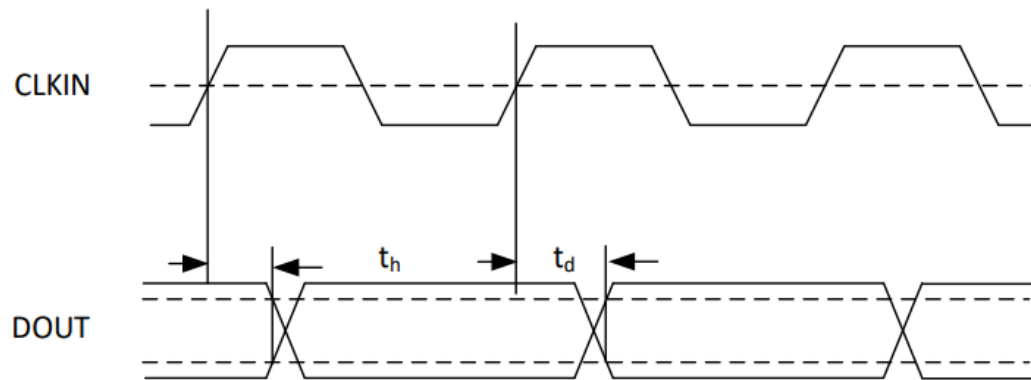


Figure 1-6. Digital Input and Output Timing

1.7 Typical Characteristics

Unless otherwise noted, test at AVDD = 5V, DVDD = 3.3V, Vin = -50mV to 50mV (CMT1306M05), CLKIN=20MHz, and sinc3 filter with OSR=256.

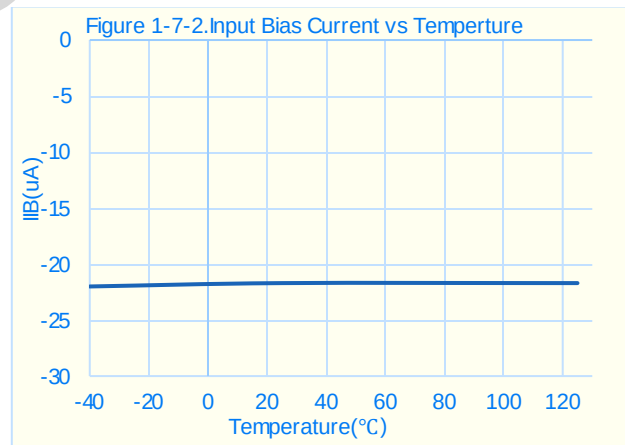
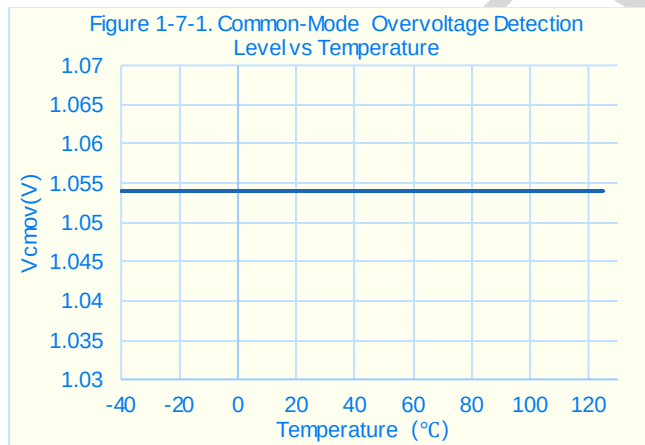


Figure 1-7-3.Common-Mode Rejection Ratio vs Temperature

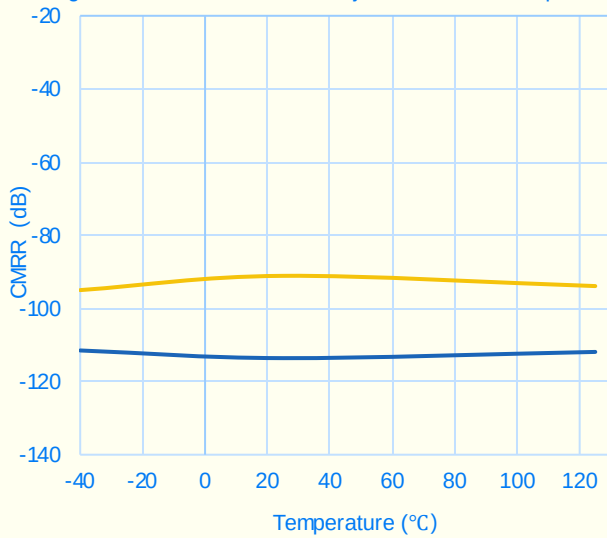


Figure 1-7-4.Input Offset vs Temperature

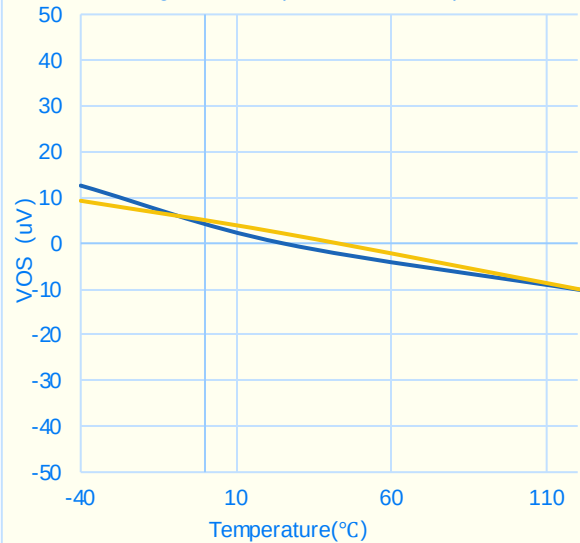


Figure 1-7-5.Input offset voltage vs Clock Frequency

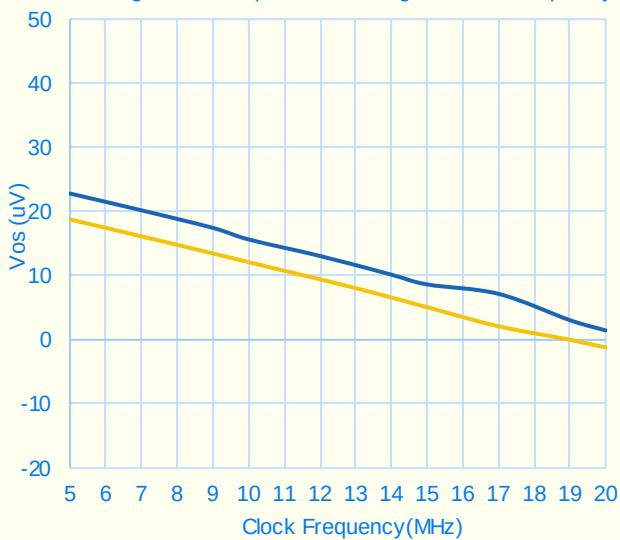


Figure 1-7-6.Gain Error vs Temperature

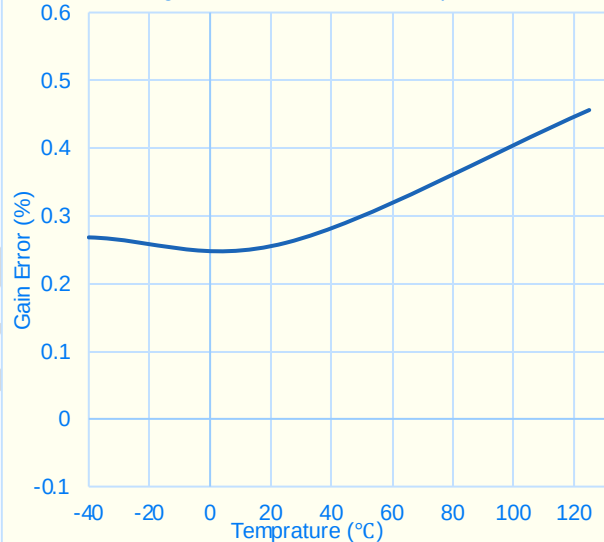


Figure 1-7-7.Gain Error vs Clock Frequency

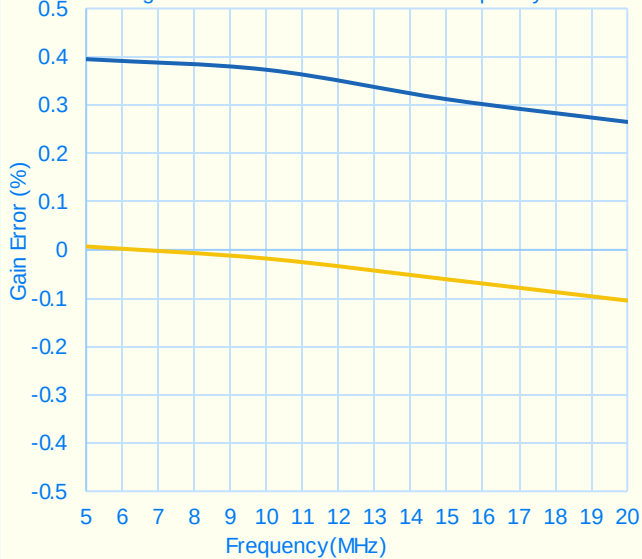


Figure 1-7-8.Signal-to-Noise Ratio vs Analog Side Supply Voltage

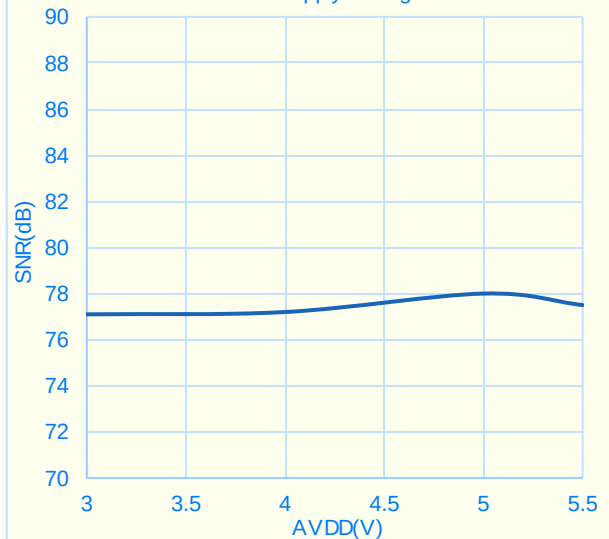


Figure 1-7-9. Signal-to-noise Ratio vs clock Frequency

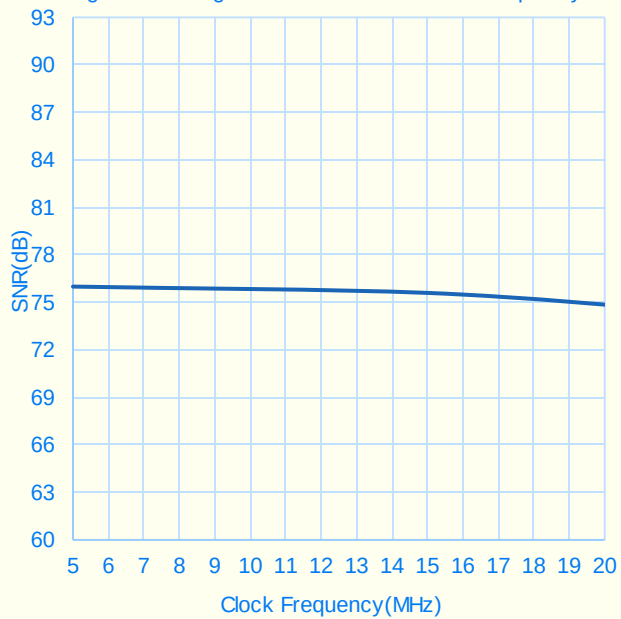


Figure 1-7-10. Signal-to-Noise Ratio vs Temperature

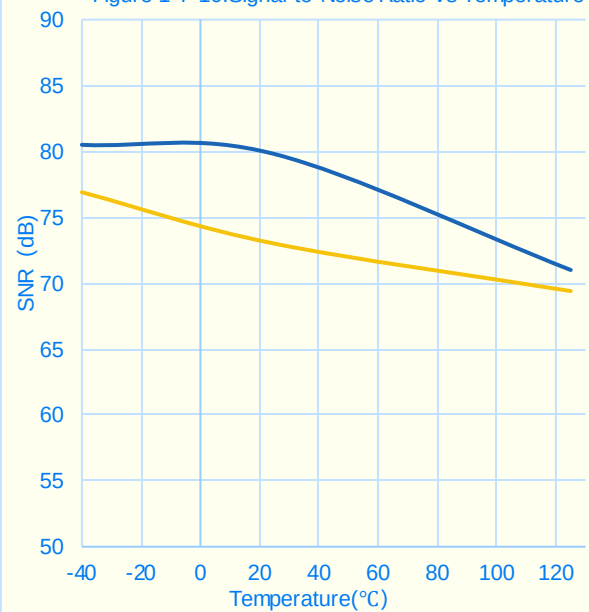


Figure 1-7-11. Total Harmonic Distortion vs Analog Side Supply Voltage

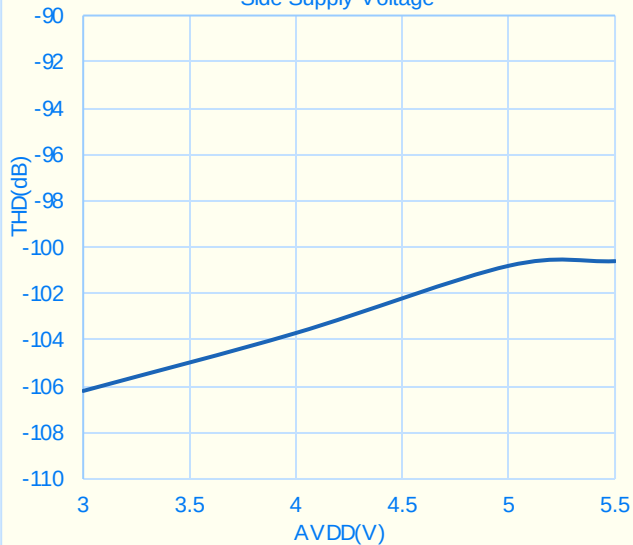


Figure 1-7-12. Total Harmonic Distortion VS Clock Frequency

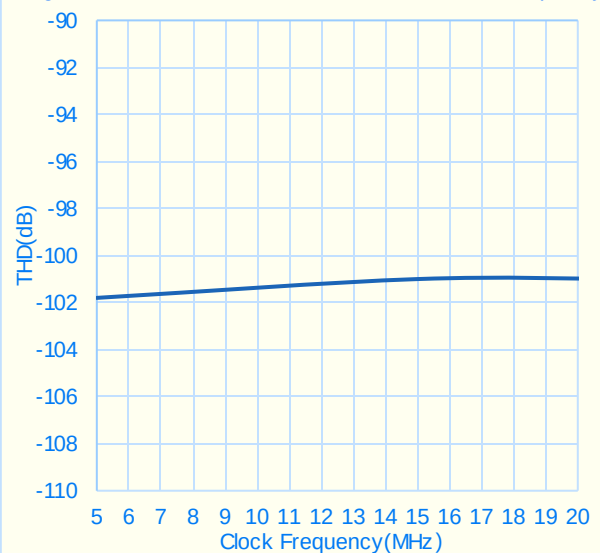


Figure 1-7-13. Total Harmonic vs Temperature

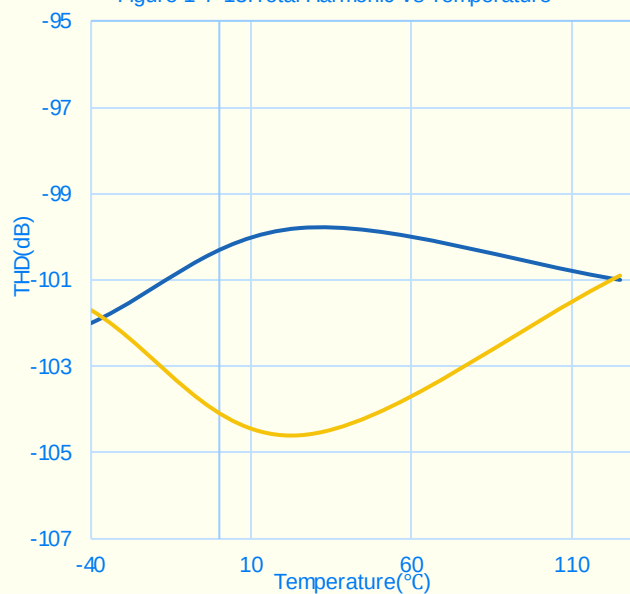


Figure 1-7-14. Signal-to-Noise + Distortion vs Analog Side Supply Voltage

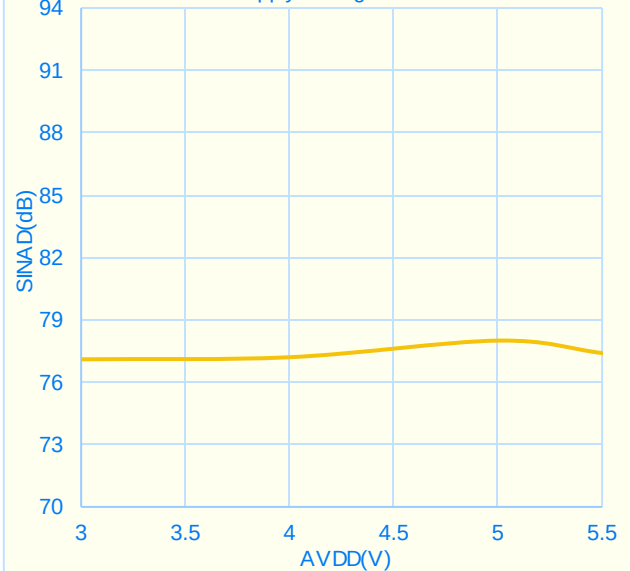


Figure 1-7-15. Signal-to-noise Distortion vs clock Frequency

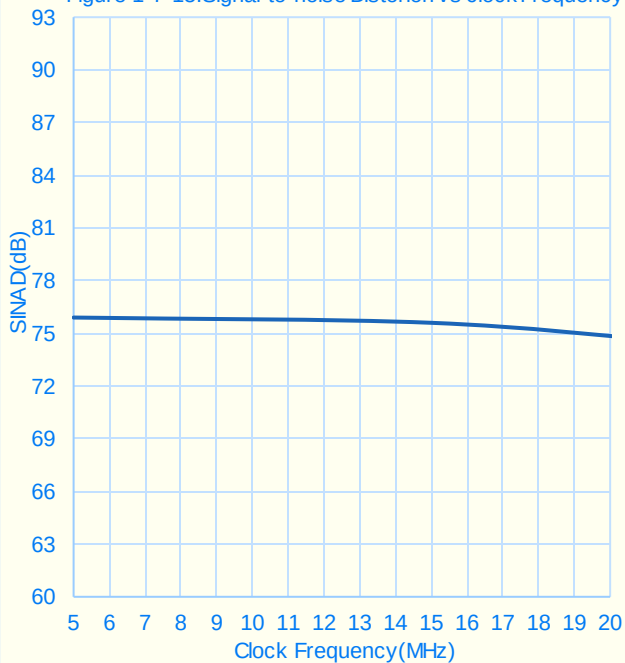


Figure 1-7-16. Signal-to-Noise + Distortion vs Temperature

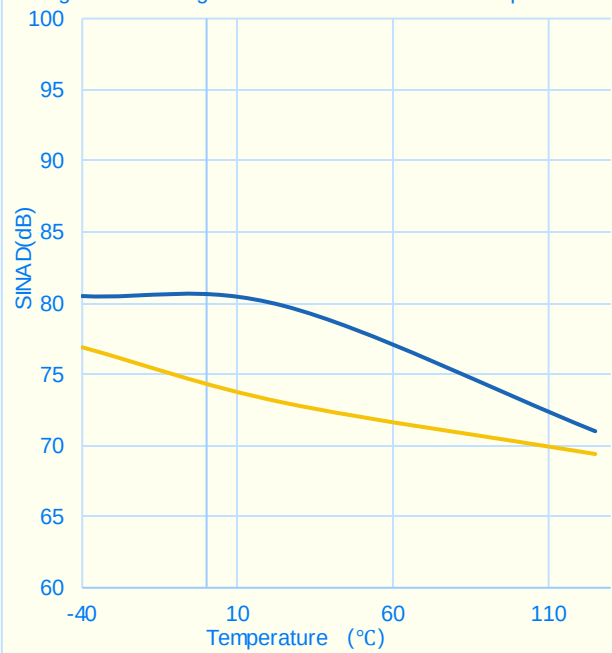


Figure 1-7-17. Spurious-Free Dynamic Range vs Analog Side Supply Voltage

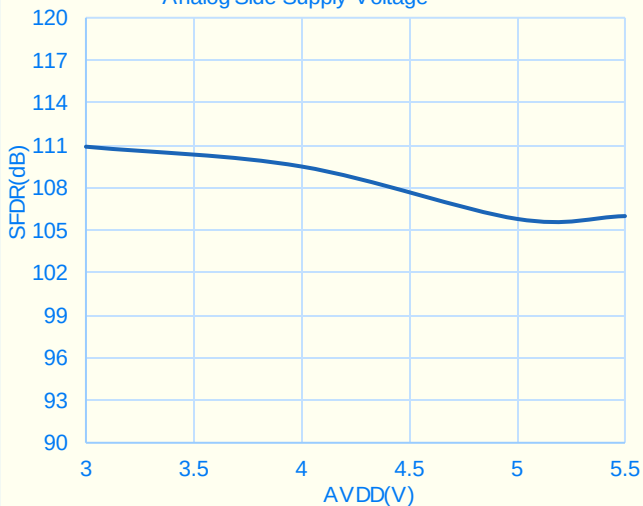


Figure 1-7-18. Spurious-Free Dange vs Clock Frequency

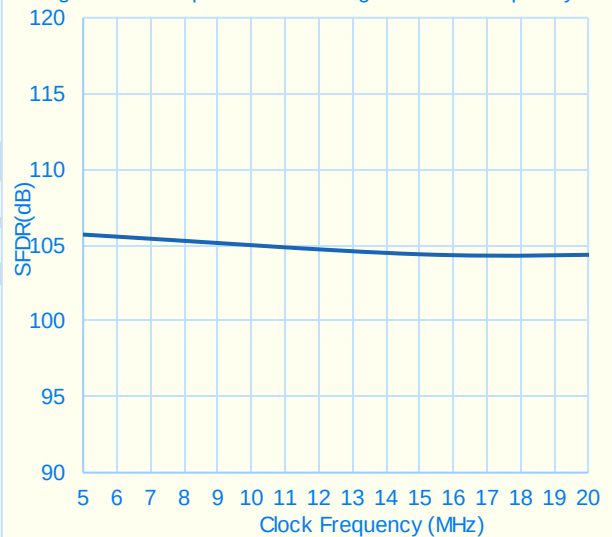


Figure 1-7-21. Analog Side Supply Current vs Supply Voltage

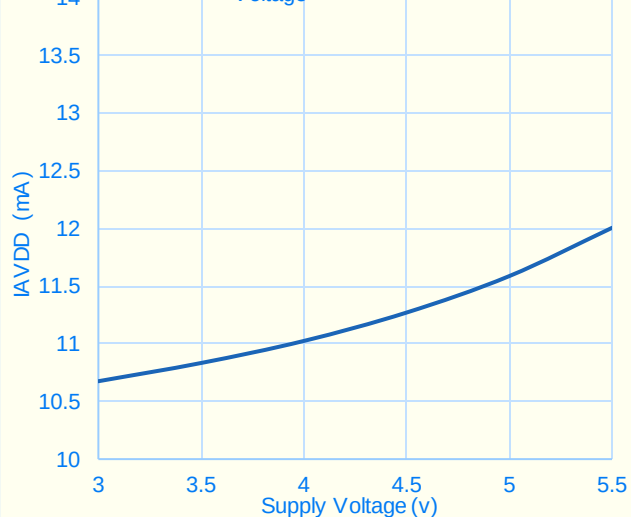


Figure 1-7-22. Analog Side Supply Current vs Clock Frequency

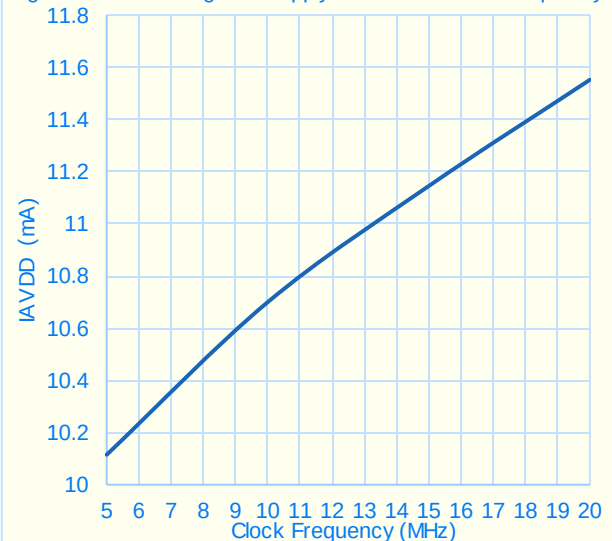


Figure 1-7-23. Digital Side Supply Current vs Clock Frequency

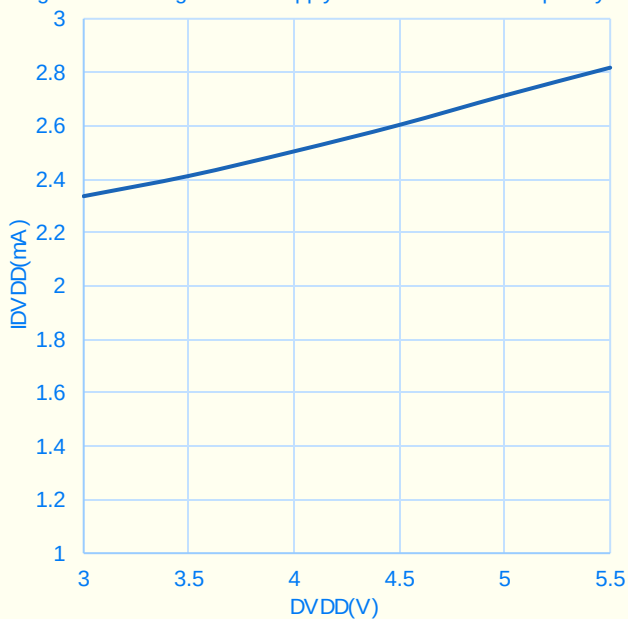


Figure 1-7-24. Digital Side Supply Current vs Clock Frequency

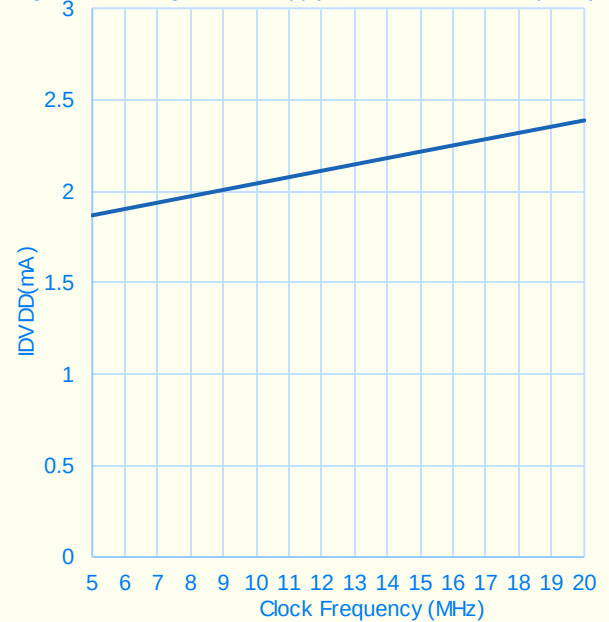


Figure 1-7-25. Supply Current vs Temperature

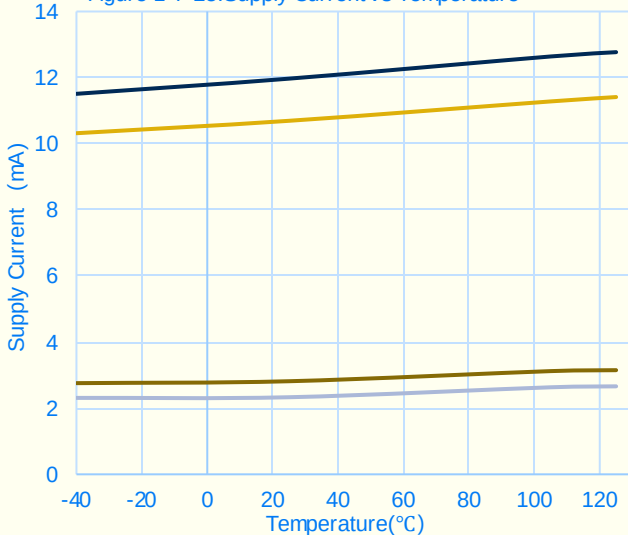


Figure 1-7-26. INL (LSB)



1.8 Safety-Related Certifications

Table 1-8. Safety-Related Certifications

VDE	CSA	UL	CQC	TUV
DIN VDE V0884-11:2017-01 (Patents pending)	IEC 60950-1, IEC 62368-1 and IEC 61010-1 (Patents pending)	Recognized under UL 1577 Component Recognition Program (Patents pending)	GB 4943.1-2011	EN 61010-1:2010 (3rd Ed) and EN 60950-1:2006/A2: 2013 (Patents pending)
Certificate number: pending	Master contract number: pending	File number: pending	Certificate number: CQC11-471543-2022	Client ID number: pending

2 Pin Description

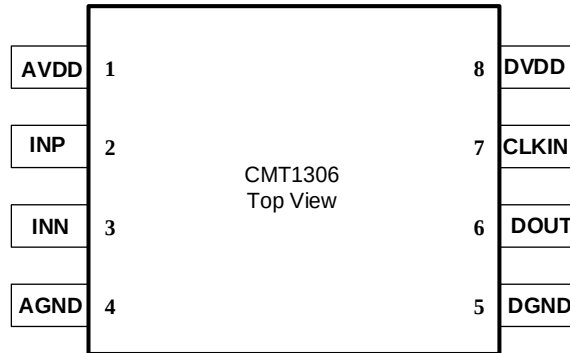


Table 2-1. CMT1306M05 SOW8 Pin Description

PIN NO.	SYMBOL	FUNCTION
1	AVDD	Power supply for analog side (3.0V to 5.5V)
2	INP	Positive analog input ($\pm 50\text{mV}$ recommended for CMT1306M05)
3	INN	Negative analog input
4	AGND	Analog ground reference
5	DGND	Digital ground reference
6	DOUT	Modulator data output
7	CLKIN	Modulator clock input: 5~21MHz
8	DVDD	Power supply for digital side (3.0V to 5.5V)

3 Function Description

3.1 Overview

The CMT1306M05 is a high performance isolated amplifier that accept fully-differential input. The fully-differential input is ideally suited to shunt current monitoring in high voltage applications where isolation is required. The analog input is continuously sampled by a second-order Σ - Δ modulator in the device, which is driven by a pre-stage fully-differential amplifier in the device. With the internal voltage reference and clock generator, the modulator converts the analog input signal to a digital bitstream. The drivers (called TX in the Functional Block Diagram) transfer the output of the modulator across the isolation barrier that separate external clock, as shown in the Functional Block Diagram.

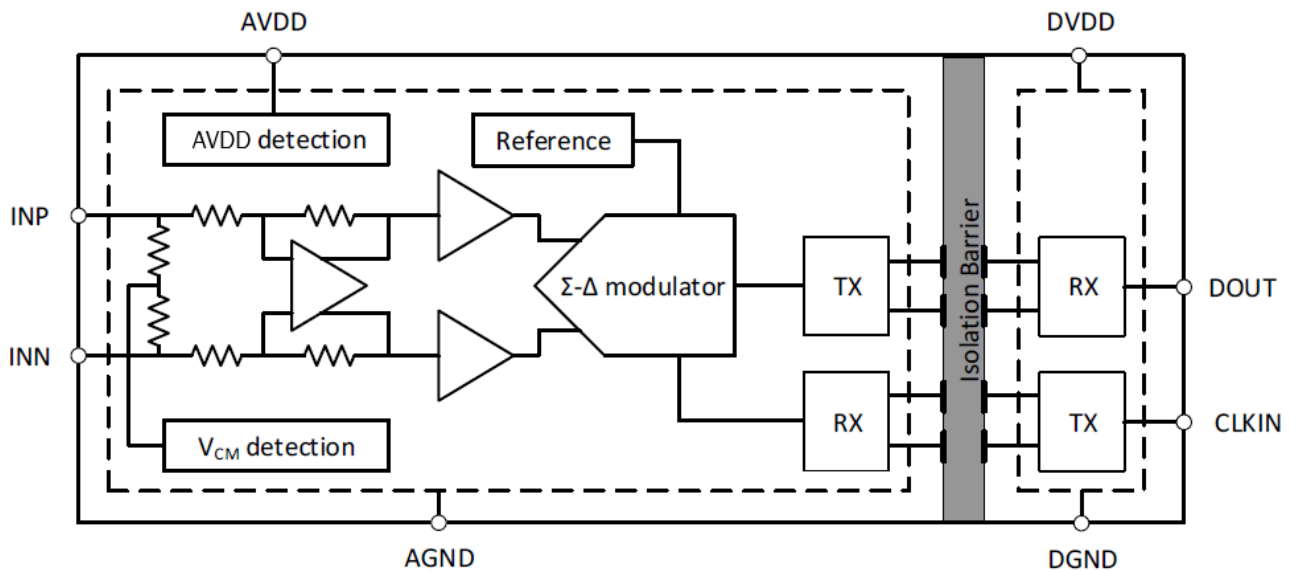


Figure 3-1. Function Block Diagram

3.2 Analog Input

There are two restrictions on the analog input signals (V_{INP} and V_{INN}).

- If the input voltage exceeds the range $AGND - 6V$ to $AVDD + 0.5V$, the input current must be limited to 10 mA because the device input electrostatic discharge (ESD) diodes turn on.
- The linearity and noise performance of the device are ensured only when the analog input voltage remains within the specified linear full-scale range (FSR) and within the specified common-mode input voltage range.

3.3 Digital Input

The digital input refers to clock signal which provides the clock for modulator conversion and output data frame clock. The clock signal should be supplied by an external source with a frequency range from 5MHz to 21MHz.

3.4 Digital Output

The digital output provides a stream of ones and zeros that can accurately represents the analog input voltage. Within the linear input range, the density of ones in the bitstream is proportional to the input voltage.

Ideally for a 0V input signal, the modulator outputs a bitstream with 50% high time. For a 250mV input signal (for the CMT1306M05), the modulator outputs a bitstream with 89.06% high time. For a -250mV input signal (for the CMT1306M05), the modulator outputs a bitstream with 10.94% high time.

If the input signal is greater than or equal to 320mV (64 mV for the CMT1306M05), the modulator clips with a steam of all ones. If the input signal is less than or equal to -320mV (-64 mV for the CMT1306M05), the output of the modulator clips with a stream of all zeros. In this case, however, the CMT1306M05 generates a single 0 (if the input is at positive full-scale) or 1 every 128 clock cycles to indicate proper device function.

Due to quantization noise of internal Σ - Δ modulator, the digital output frequency is not fixed, but the density of high time is fixed based on input voltage. Typical modulator digital output characteristics are as below.

Table 3-4. Input voltage with ideal corresponding density of high time at modulator data output and ADC code

Analog input	Input voltage	Density of high time	Digital output code (16-bit unsigned decimation)
+Differential input voltage before clipping output	+320mV (+64mV for CMT1306M05)	99.22%	65024
+Linear differential input full scale voltage	+250mV (+50mV for CMT1306M05)	89.06%	58366
Zero	0mV	50%	32768

Ideal density of 1s in the output bitstream for any input voltage (except for full-scale input, as described in section 2.5) at modulator data output can be calculated with:

$$\frac{V_{IN} + V_{CLIPPING}}{2 \times V_{CLIPPING}}$$

Similarly, the ADC code can be calculated, assuming a 16-bit unsigned decimation filter.

3.5 Fail-safe Output

CMT1306M05 integrates some diagnostic measures and offers a fail-safe output to simplify system-level design. The fail-safe function will be activated in the following conditions:

- When the undervoltage of AVDD is detected ($AVDD < AVDD_{UV}$), DOUT pin output a bitstream of all logic zeros, as shown in Figure 4.5.
- When the overvoltage of common-mode input voltage is detected ($V_{CM} > V_{CMOV}$), DOUT pin output a bitstream of all logic ones, as shown in Figure 4.5

NOTE: If both of the faults above occur at the same time, DOUT pin output a bitstream of all logic zeros. (AVDD missing has a higher priority).

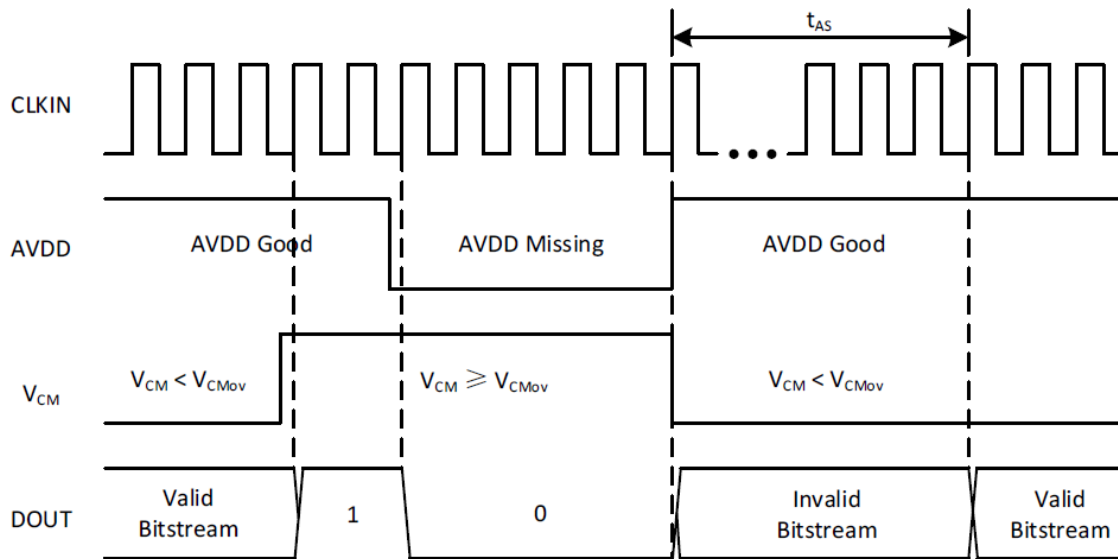


Figure 3-5. Fail-safe output

If an overrange input signal is applied to the CMT1306M05 ($V_{IN} \geq V_{Clipping}$), the output generates a single 0 or 1 every 128 clock cycles, as shown in Figure 3-6.

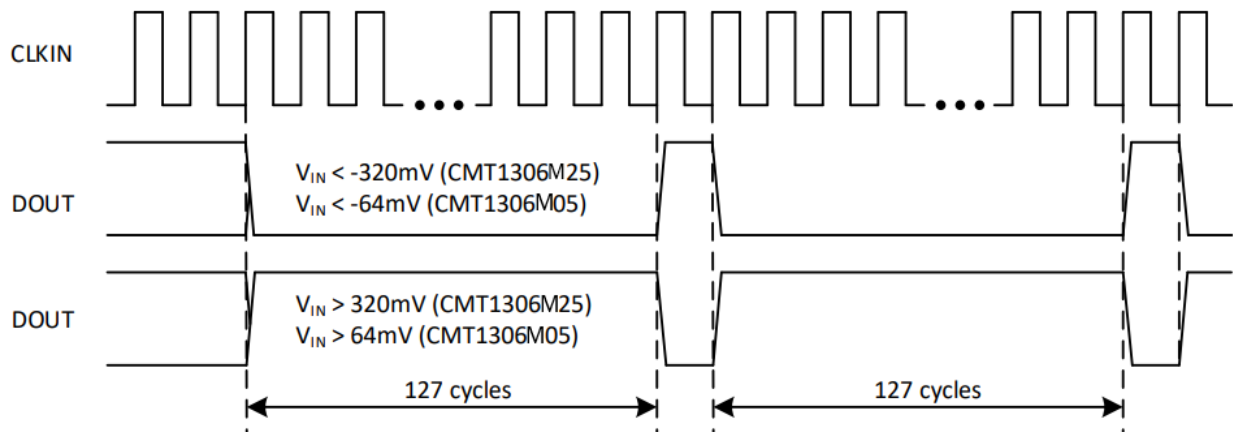


Figure 3-6. Overage Output

4 Application Note

4.1 Typical Application Circuit

CMT1306M05 is ideally suited to shunt resistor-based current sensing in high voltage applications such as frequency inverters. The typical application circuit is shown in Figure 4.1.

The voltage across the shunt resistor R_{sense} is applied to the differential input of CMT1306M05 through a RC filter. The internal

second-order sigma-delta modulator converts the analog input to a single-bit output stream. The external digital system provides a clock source for the modulator and a digital filter for decimation and quantization noise filtering.

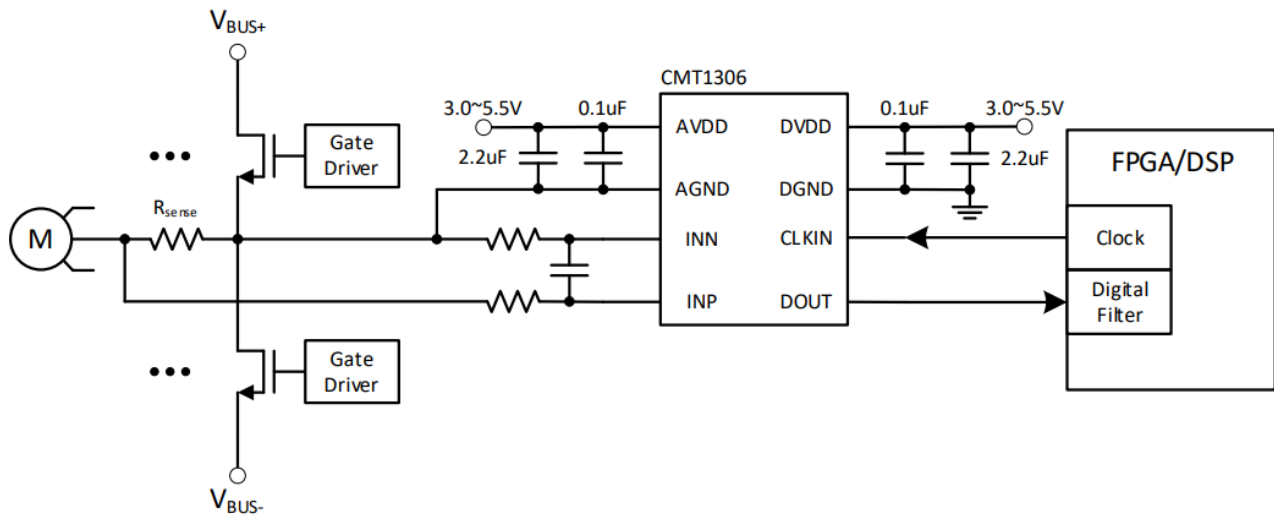


Figure 4-1. Typical Application Circuit in Phase Current Sensing

4.2 Shunt Resistor Selection

Choosing a particular shunt resistor is usually a compromise between minimizing power dissipation and maximizing accuracy. Smaller sense resistor decreases power dissipation, while larger sense resistor can improve measure accuracy by utilizing the full input range of isolated amplifier.

There are two other factors should be considered when selecting the shunt resistor:

- The voltage-drop caused by the rated current range must not exceed the recommended linear input voltage range:
 $V_{SHUNT} \leq FSR.$
- The voltage-drop caused by the maximum allowed overcurrent must not exceed the input voltage that causes a clipping output: $V_{SHUNT} \leq V_{Clipping}.$

4.3 Digital Filter

The Σ - Δ modulator a characteristic of noise shaping. Most of the quantization noise is pushed from a low frequency to a higher frequency.

In order to reduce higher-frequency quantization noise, the modulator output is fed to the digital low-pass filter. Subsequently, the signal of interest passes through to the output of the digital filter, while much of the higher-frequency quantization noise is filtered out.

The digital filter serves another function – decimation. It creates a digital output code from the bitstream that the modulator outputs. The ratio of the modulator rate (f_{MOD}) of the delta-sigma modulator to its output data rate (f_{DR}) is the oversampling ratio (OSR). The relationship between f_{DR} and f_{MOD} is:

$$f_{DR} = f_{MOD} / OSR \quad \text{Equation 4.1}$$

A sinc3 filter is recommended since it's simple and requires less hardware resources. Equation 4.2 describes the transfer function of a sinc filter.

$$H(Z) = \left(\frac{1}{DR} \frac{(1-Z^{-DR})}{(1-Z^{-1})} \right)^N$$

Equation 4.2

where:

DR is the decimation rate;

N is the sinc filter order.

The filter can be implemented in an FPGA or DSP. The sinc filter creates a digital output code by taking a multi-order moving average of the modulator output over a certain number of modulator clock periods.

The higher the decimation rate, the higher the conversion accuracy, and the lower the output data rate. So, there is a trade-off between accuracy and data rate. All the characterization in this datasheet is tested with a sinc3 filter with an oversampling ratio (OSR) of 256.

The output data size is expressed in Equation 4.3. The 16 most significant bits are used to return a 16-bit result.

$$Data\ Size = N \times \log_2 DR$$

Equation 4.3

The filter characteristics for a third-order sinc filter are summarized in Table 4.1.

Table 4-1. Sinc3 Filter Characteristics for 20 MHz CLKIN

Decimation Rate (DR)	Data Output Rate (kHz)	Data Size (bits)	Filter Response (kHz)
32	625	15	163.7
64	312.5	18	81.8
128	156.2	21	40.9
256	78.1	24	20.4
512	39.1	27	10.2

4.4 PCB Layout

There are some key guidelines or considerations for optimizing performance in PCB layout:

- CMT1306M05 requires a 0.1μF bypass capacitor between AVDD and AGND, DVDD and DGND. The capacitor should be placed as close as possible to the VDD pin. If better filtering is required, an additional 1~10μF capacitor may be used.
- Kelvin rule is recommended for the connection between shunt resistor to CMT1306M05. Because of the Kelvin connection, any voltage drops across the trace and leads should have no impact on the measured voltage.
- Place the shunt resistor close to the INP and INN inputs and keep the layout of both connections symmetrical and run very close to each other to the input of the CMT1306M05. This minimizes the loop area of the connection and reduces the possibility of stray magnetic fields from interfering with the measured signal.

5 Ordering Information

Table 5-1. CMT1306M05 Ordering Information

Part Number	Isolation Rating (V _{RMS})	Operating Condition	Auto-motive	Pakcage	Minimum Order Quantity
CMT1306M05	5000	-40 to 125°C	/	SOW8	1000

Please visit www.hoperf.com for more product/product line information.

Please contact sales@hoperf.com or your local sales representative for sales or pricing requirements.

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6 Packaging Information

The packaging information of the CMT1306M05 is shown in the figure below.

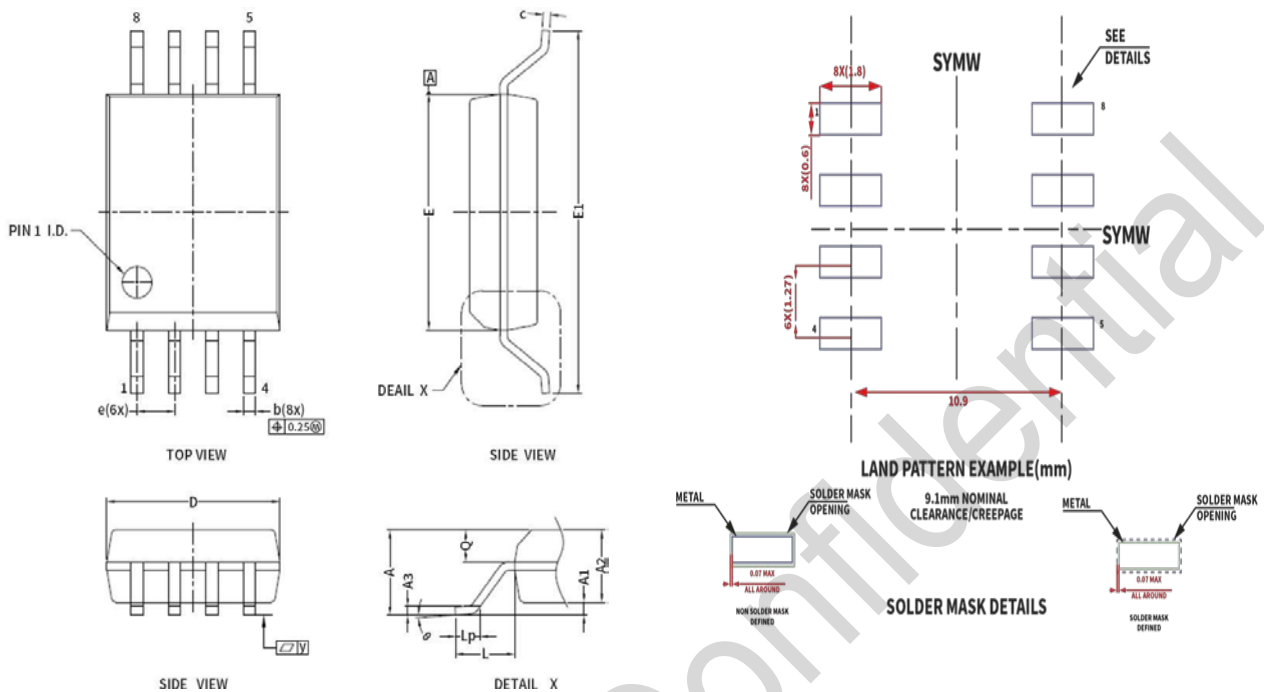


Figure 6-1.CMT1306M05 SOW 8 Packaging

Table 6-1. CMT1306M05 SOW 8 Packaging Scale

Symbol	Scale (mm)		
	Min.	Typ.	Max.
A	-	-	2.80
A1	0.36	-	0.46
A2	2.20	2.30	2.40
A3	-	0.25	-
Q	0.97	1.02	1.07
b	0.31	0.41	0.51
c	0.13	-	0.33
D	5.75	5.85	5.95
E	7.40	7.50	7.60
E1	11.25	11.50	11.75
e	1.27 bsc		
L	2.00 bsc		
Lp	0.50	-	1.00
Y	-	0.10	-
θ	0	-	8°

7 Tape and Reel Information

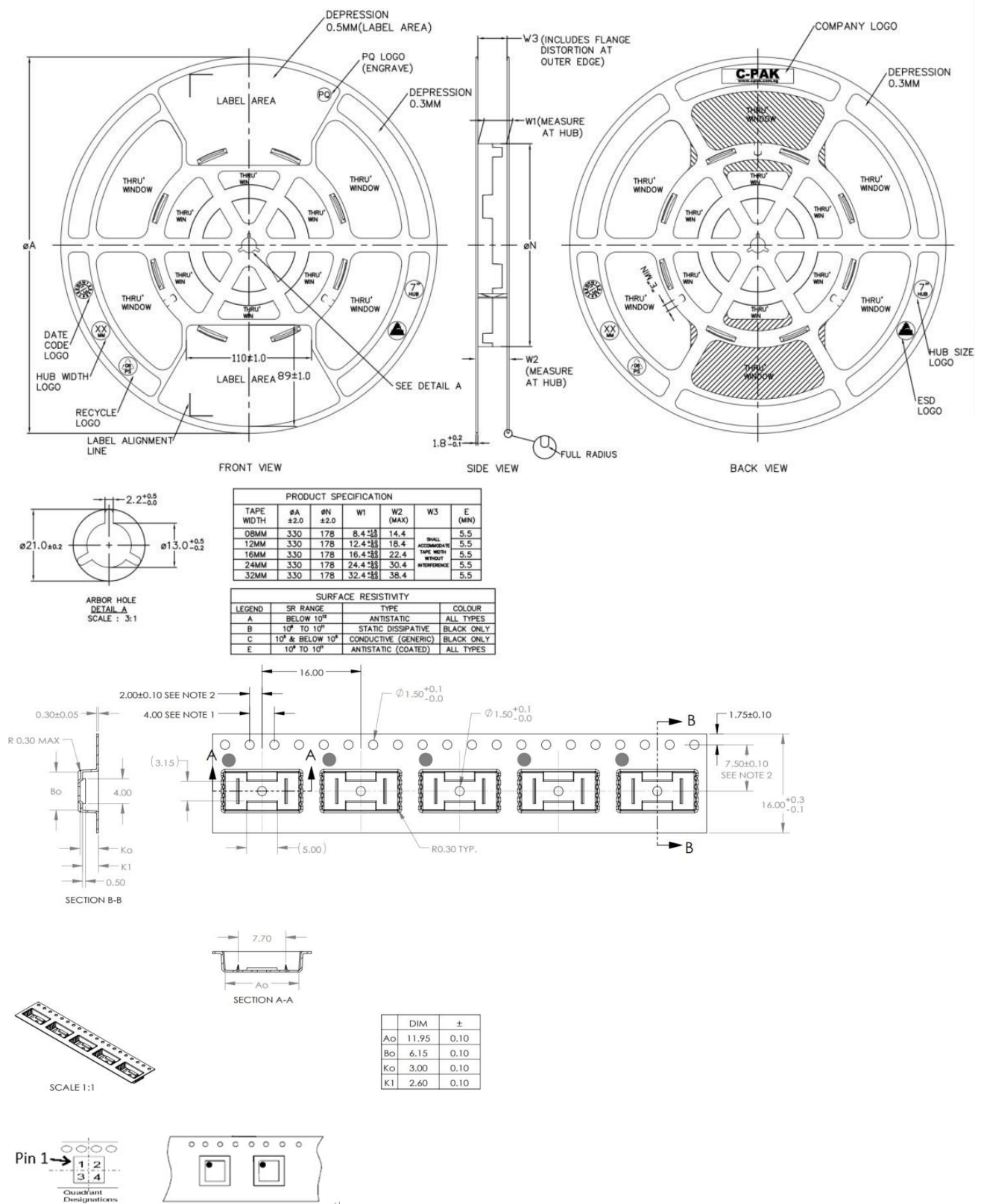


Figure 7-1. Tape and Reel Information of CMT1306M05 SOW8(300mil)

8 Revise History

Table 8-1. Revise History Records

Version No.	Chapter	Description	Date
0.1	All	Initial version	2024-06-14

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9 Contacts

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